

Question 2: Natural Deduction Proofs

a) $\forall x(P(x) \rightarrow Q(x)), \exists xP(x) \vdash \exists xQ(x)$

1.	$\forall x(P(x) \rightarrow Q(x))$	premise
2.	$\exists xP(x)$	premise
3.	$x_0 \quad P(x_0)$	assume
4.	$P(x_0) \rightarrow Q(x_0)$	$\forall e \ 1$
5.	$Q(x_0)$	$\rightarrow e \ 4, 3$
6.	$\exists xQ(x)$	$\exists i \ 5$
7.	$\exists xQ(x)$	$\exists e \ 2, 3-6$

b) $\forall xP(x) \vee \forall xQ(x) \vdash \forall x(P(x) \vee Q(x))$

1.	$\forall xP(x) \vee \forall xQ(x)$	premise
2.	$\forall xP(x)$	assume
3.	x_0	arbitrary
4.	$P(x_0)$	$\forall e \ 2$
5.	$P(x_0) \vee Q(x_0)$	$\vee i_1 \ 4$
6.	$\forall x(P(x) \vee Q(x))$	$\forall i \ 3-5$
7.	$\forall xQ(x)$	assume
8.	y_0	arbitrary
9.	$Q(y_0)$	$\forall e \ 7$
10.	$P(y_0) \vee Q(y_0)$	$\vee i_2 \ 9$
11.	$\forall x(P(x) \vee Q(x))$	$\forall i \ 8-10$
12.	$\forall x(P(x) \vee Q(x))$	$\vee e \ 1, 2-6, 7-11$

Question 3: Model Finding

a) $\forall x \exists y P(f(x), y)$

Model that satisfies ($\mathcal{M} \models \phi$):

- Domain $D = \mathbb{Z}$ (integers).
- $f(x) = x$ (identity function).
- $P(x, y)$ is defined as the relation $x < y$.
- **Explanation:** For every integer x , there exists an integer y (such as $y = x + 1$) such that $x < y$ holds.

Model that does not satisfy ($\mathcal{M} \not\models \phi$):

- Domain $D = \{1, 2\}$.
- $f(x) = 1$ for all x .
- $P(x, y)$ is always *false*.
- **Explanation:** Since $P(1, 1)$ and $P(1, 2)$ are both false, there is no y that satisfies the relation for any x .

b) $(\forall x P(x) \rightarrow S) \vdash \exists x (P(x) \rightarrow S)$

Model that satisfies:

- Domain $D = \{1\}$.
- $P(1) = \text{true}$.
- $S = \text{true}$.
- **Explanation:** The premise $(\forall x P(x) \rightarrow S)$ is $\text{true} \rightarrow \text{true}$, which is true. The conclusion $\exists x (P(x) \rightarrow S)$ is true because $P(1) \rightarrow S$ is true.

Model that does not satisfy (Counterexample):

- Domain $D = \emptyset$ (The Empty Set).
- **Explanation:** In standard first-order logic, the domain must be non-empty. However, if we consider a non-empty domain $D = \{1\}$ where:
- $P(1) = \text{false}$ and $S = \text{false}$.
- The premise $(\forall x P(x) \rightarrow S)$ becomes $(\text{false} \rightarrow \text{false})$, which is **true**.
- The conclusion $\exists x (P(x) \rightarrow S)$ becomes $(\text{false} \rightarrow \text{false})$, which is also **true**.
- *Note:* This specific sequent is actually a valid logical principle in classical logic with non-empty domains, meaning a counterexample would only exist if the domain were empty, which is usually disallowed.